



DELIVERY ROUTE OPTIMIZER

Data Structures and Algorithms – III

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Section 10 • Guided by Ms. K. Chandusha

1. Introduction

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Overview

Delivery route optimization aims to plan efficient routes for multiple vehicles.



The Challenge

Inefficient route planning can increase travel distance, fuel consumption and delivery time.



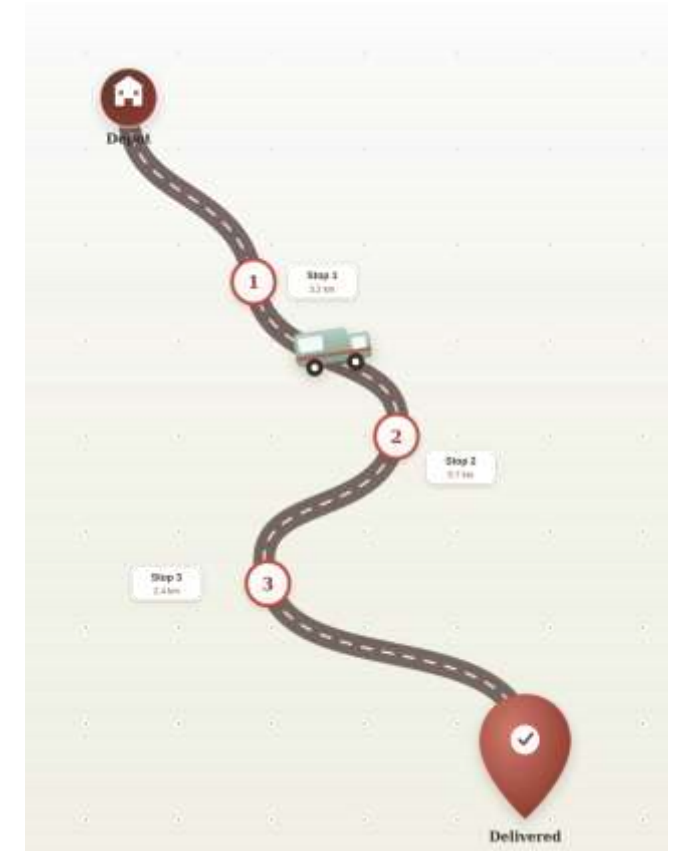
The Approach

The proposed system generates near-optimal delivery routes, considering delivery locations, vehicle capacity and customer time windows.



The Goal

Improve vehicle utilization, reduce operational cost and support timely deliveries.



A delivery route: depot → stops → destination

2. Abstract

02

- Generates near-optimal routes for multiple delivery vehicles rather than relying on manual planning.
- Considers delivery locations, vehicle capacity and customer time-window constraints together.
- Cleans and standardizes input data (stops, vehicle details) before optimization begins.
- Applies graph-based modeling with route optimization, dynamic programming and approximation/heuristic techniques as applicable.
- Uses approximation/heuristic methods when exact optimization becomes computationally expensive for larger instances.
- Reports route-quality measures — distance, travel time, fuel usage and constraint satisfaction — for evaluation.



3. Software Requirements

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Programming Environment

A suitable programming environment for implementing the routing algorithms.



Data Structures

Graph-based representation of delivery locations and routes.



Input Data

Delivery stops, vehicle information, capacities and customer constraints.



Operating System

Windows or Linux.



Algorithms

Route optimization, dynamic programming and approximation/heuristic techniques as applicable.

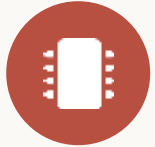


Output

Optimized routes and route-performance measures.

4. Hardware Requirements

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Processor

Intel Core i5 or equivalent.



RAM

8 GB or higher.



Storage

256 GB or higher.



Display

Standard laptop/desktop display.



Network

Internet connection when external map or location data is required.

Note: Requirements target a typical development laptop — no specialized hardware is needed.

5. Literature Survey

Approach / Problem	Technique	Key Limitation / Observation
Travelling Salesperson Problem (TSP)	Route optimization	Exact solutions become expensive as the number of locations increases.
Vehicle Routing Problem (VRP)	Multi-vehicle route planning	Must handle multiple vehicles and practical constraints.
Dynamic Programming	Optimal subproblem-based computation	Computational cost can grow rapidly for larger routing instances.
Approximation Algorithms	Near-optimal optimization	Trades exact optimality for practical computation time.
Proposed Delivery Route Optimizer	Optimization with practical constraints	Targets efficient routes while considering capacity, locations and time windows.

6. Methodology

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1

Data Collection

Gather delivery locations, vehicle details and customer constraints.

2

Data Preprocessing

Clean and standardize delivery and stop information.

3

Vehicle Assignment

Assign delivery stops based on capacity, location and time constraints.

4

Route Generation

Construct feasible routes between delivery points.

5

Route Optimization

Reduce total travel distance, estimated time and fuel consumption.

6

Constraint Checking

Verify vehicle capacity and customer time-window requirements.

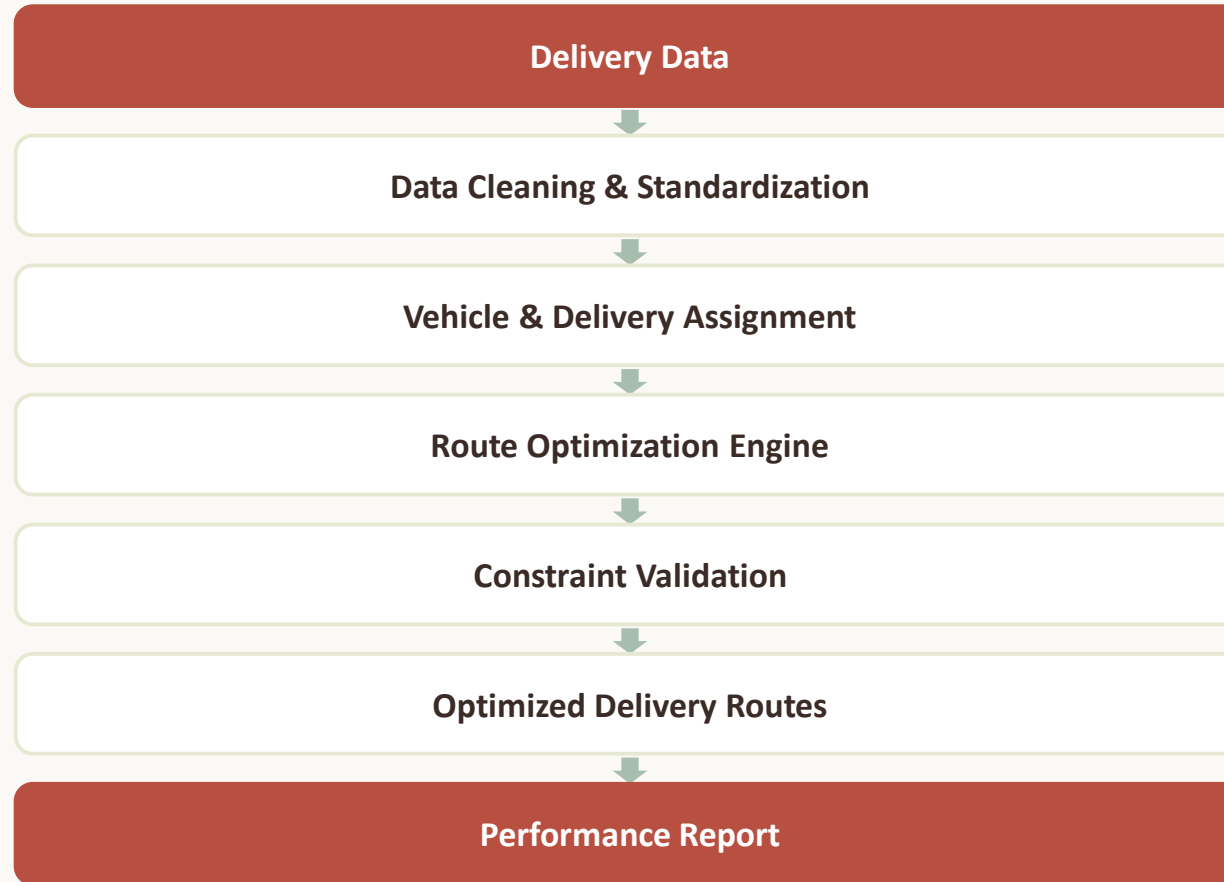
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Evaluation

Report distance, travel time, fuel usage, route efficiency and constraint satisfaction.

Methodology – Overall Flow

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Conclusion



The Delivery Route Optimizer provides a practical approach to improving delivery planning.



It aims to reduce travel distance, fuel consumption and delivery time.



Near-optimal optimization methods make the approach practical for larger routing problems.



The system can support cost-efficient and timely delivery operations.

THANK YOU • QUESTIONS?